Brandon Jahoor

I give robots the ability to perceive, decide, and act

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University of Waterloo | *Candidate for BASc in Mechatronics Engineering Honours, Minor in Artificial Intelligence* *2028*

Skills

Robotics: Isaac Sim & Lab, URDF/USD, ROS2, controls, IK/FK, Nav2, path planning
ML & Perception: ACT, VLA, RL, LeRobot, Hugging Face, computer vision, OpenCV, RealSense
Software: Python, C++, Docker, Linux/Ubuntu, Claude Code, llama.cpp, Ollama
Hardware & Design: SolidWorks, 3D printing, SO-ARM101, Nvidia Jetson, Raspberry Pi, prototyping

Experience

Robotics Engineering (Co-op) — Jan 2026 – April 2026
Thalassa Robotics | *site* · *portfolio* — *California*

- Built a high-fidelity toolchain on **Isaac Sim** used team-wide, integrating and configuring **optics and hydrodynamics** it lacks.
- Prepared the robot's **USD** model and joint articulation so the simulated assembly matched the real one.
- Layered control abstractions so simulated motors drive exactly like the real servos, with acceleration-bounded interpolation.

Robotics Research Assistant — Sept 2025 – Dec 2025
University of Waterloo RoboHub | *gitlab* · *site* · *portfolio* — *Ontario*

- Streamed **human arm motion** from **Xsens IMUs** into **ROS2** in real time, for retargeting onto a 7-axis **Franka Emika Panda**.
- Containerised the **ROS2** stack in **Docker** so results reproduced across the lab's workstations.

Robotics Engineering (Co-op) — June 2025 – Aug 2025
University of Waterloo Robotics Team | *github* · *site* · *portfolio* — *Ontario*

- Developed the **ROS2 RealSense** RGB-D stream with custom compression and dynamic resolution for a bandwidth-limited link.
- Modelled a 6-wheel rocker-bogie drivetrain and solved its **forward/inverse kinematics** for a Mars rover on an **Nvidia Jetson**.

Robotics Engineering (Co-op) — May 2025 – June 2025
BHF Robotics | *site* · *portfolio* — *Ontario*

- Implemented **ROS2 autodrive** and trained a **deep-learning** weed/crop detector for an autonomous farming robot.
- Designed, **FEA**-validated and installed parts, then debugged and improved its mechanical, electrical and software systems.

Projects

VLA Finetuning | *SmolVLA, LeRobot, SO-ARM101, Nvidia Jetson* | github · portfolio

- Trained **ACT** and **SmolVLA** on 50–100 teleoperated demos/task, running **30 Hz** closed-loop on a headless Jetson Orin Nano.
- Profiled training on an RTX 3060 Ti: GPU-bound at a **30:1** compute-to-dataloader ratio, pinning batch and worker counts.

Chess Robot | *SO-ARM101, SolidWorks, 3D printing, LeRobot, Placo, Python* | *Stealth Startup* | portfolio

- **CAD**-modified an **SO-ARM101** and built its rig, then developed a three-call Python API that plugs straight into a chess engine.
- Extended reach and fused the wrist to drop a motor, updating the **URDF** model, solver constraints and **LeRobot** driver.
- Built the motion engine over **LeRobot** and **Placo** IK: convergence loops, **1 mm** waypoints, **PID gravity compensation**.

Autonomous Navigation Stack | *ROS2, C++, LiDAR, A*, Foxglove* | *WATonomous* | github · portfolio

- Wrote a four-node pipeline: **LiDAR** costmap, global map, **A*** planner, Pure Pursuit control.

Robot Arm Simulation | *Isaac Sim, Isaac Lab, CUDA, RL* | *Stealth Startup* | portfolio

- Trained RL control policies for Franka and SO-ARM101 arms in CUDA-enabled **Isaac Lab**; validated sim-to-real readiness.

Hexapod Robot | *Nvidia Jetson, OWL-ViT, SO-ARM101, RealSense* | portfolio

- Led zero-shot **OWL-ViT** integration on an 18-DOF hexapod, letting it follow any object named in a text prompt.

AI Roommate | *Nvidia Jetson, Qwen2.5-0.5B, YOLOv8s, FastAPI* | github · portfolio

- Grounded a local LLM in live **YOLOv8s** detections so it answers questions about what the camera sees, on-device.

Autonomous Rover Simulation | *Gazebo, ROS2, Nav2, Python* | *RoboHub* | github · portfolio

- Developed a **ROS2/Nav2** navigation stack with six-sector obstacle avoidance and plant classification for a **Gazebo** rover.

AI Object Detection | *ROS2, OWL-ViT, RealSense, OpenCV, Python* | github · portfolio

- **ROS2** package pairing compressed RealSense RGB-D with **OWL-ViT**, retargeted by text on a live topic.

See my portfolio: bjahoor.github.io